



High Level Design(HLD)

Regional Sales Data Analysis

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Abstract

This project focuses on analyzing regional sales performance using historical sales data collected from multiple regions across the United States. The objective is to transform raw sales records into meaningful business insights through data cleaning, transformation, analysis, and visualization techniques. Python and Google Colab were used for data preprocessing, while Power BI was used to build interactive dashboards. The analysis helps identify revenue trends, profit performance, customer purchasing behavior, product performance, channel contribution, and regional sales distribution. The final dashboard supports data-driven decision-making by presenting key performance indicators and business insights in a clear and interactive format.

1. Introduction

1.1 Why this High-Level Design Document?

This High-Level Design document provides an overview of the Regional Sales Data Analysis project. It describes the overall workflow, tools used, data processing techniques, dashboard design, and key business insights generated from the analysis. The document serves as a reference for understanding how raw sales data is converted into meaningful visual reports that support business decision-making.

1.2 Scope

The scope of this project includes:

- Analyze revenue and profit performance.
- Study monthly sales trends.
- Identify top-selling products.
- Evaluate customer purchasing behavior.
- Compare regional and state-wise sales performance.
- Analyze sales channel contributions.

- Present findings using interactive Power BI dashboards.

2. General Description

This section provides an overview of the Regional Sales Analysis project. The project focuses on analyzing sales data from different regions, customers, products, and sales channels using Python and Power BI. The objective is to transform raw sales data into meaningful business insights through data cleaning, transformation, feature engineering, and interactive dashboard development.

The dataset used in this project was sourced from Kaggle, a widely used public platform for data science, analytics, and machine learning datasets.

2.1 Product Perspective

This project is a business analytics solution that converts raw sales data into meaningful insights using visualization and reporting tools. The dataset used for this project was obtained from Kaggle, a public data science and machine learning platform. The project helps users understand revenue trends, profit performance, customer behavior, product performance, and regional sales distribution through interactive dashboards and reports.

2.2 Problem Statement

The company conducts sales operations across multiple regions, states, products, customers, and sales channels. Due to the large volume of sales data generated over time, it becomes difficult for management to manually monitor business performance and identify important trends. Revenue and profit may vary significantly across regions, while certain products and customers may contribute more to overall business growth than others. Additionally, management lacks a clear understanding of monthly sales patterns, regional performance, channel-wise contribution, and customer purchasing behavior. Without proper analysis and visualization, valuable business opportunities and performance issues may remain unnoticed. The objective of this project is to analyze historical sales data, identify key performance indicators, and develop interactive dashboards that provide

meaningful insights into revenue, profit, customer behavior, product performance, and regional sales trends. These insights will help management make informed decisions, improve profitability, and identify future growth opportunities.

2.3 Proposed Solution

The solution involves cleaning and transforming sales data using Python and Power Query, creating business metrics such as Revenue, Profit, and Profit Margin, and developing interactive Power BI dashboards. The dashboards help users monitor business performance and support data-driven decision-making.

2.4 Data Requirements

The project requires datasets containing sales orders, customer details, product information, regional information, and budget data. The datasets used in this project were obtained from Kaggle's public dataset repository and include Sales Orders, Customers, Products, Regions, State Regions, and Budget data. The data should be accurate, properly formatted, and free from duplicates to ensure reliable analysis and reporting.

2.5 Tools Used

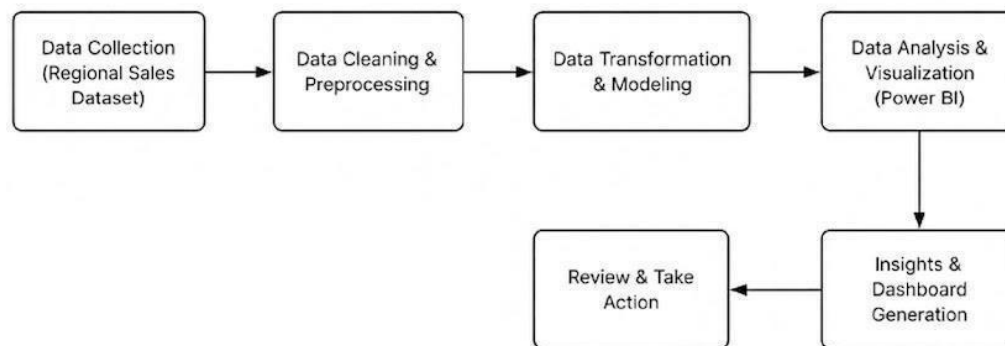
The tools used in this project include **Python, Google Colab, Microsoft Excel, Power BI, Power Query, DAX, and ChatGPT (OpenAI)**. Python and Google Colab were used for data cleaning, preprocessing, feature engineering, and exploratory data analysis. Microsoft Excel was used for dataset storage and review. Power Query was used for data transformation and preparation, while Power BI and DAX were used for data modeling, KPI creation, and dashboard development. **ChatGPT (OpenAI)** was utilized to assist with understanding concepts, documentation preparation, report writing, and improving the overall project workflow.



3. Design Details

This section explains how the data is processed step by step to convert raw information into meaningful insights using Power BI.

3.1 Process Flow



3.1.1 Data Cleaning and Transformation

Data cleaning was performed using Python in Google Colab to improve data quality and consistency. Duplicate records were removed, missing values were handled, and data types were standardized. Power Query was then used to transform the data, rename columns, merge datasets, and establish relationships between tables. These steps ensured that the data was accurate and ready for analysis.

3.1.2 Dashboard Development Process

After data preparation, the cleaned datasets were imported into Power BI. Relationships between tables were created and DAX measures were developed to calculate key business metrics such as Revenue, Profit, and Profit Margin. Various visualizations including KPI cards, line charts, bar charts, maps, and donut charts were then designed to provide meaningful business insights.

3.2 Event Log

Event logging in this project refers to tracking important stages such as data import, data cleaning, data transformation, relationship creation, and dashboard updates. These logs help ensure that the data processing workflow is executed correctly and consistently.

3.3 Error Handling

Error handling was performed during the data preparation stage. Common issues such as missing values, duplicate records, incorrect data formats, and inconsistent entries were identified and corrected using Python and Power Query. This ensured accurate calculations and reliable dashboard results.

4. Performance

The performance of this project depends on the efficiency of data cleaning, transformation, modeling, and visualization processes. By using Python for preprocessing and Power BI for visualization, the project ensures accurate analysis and smooth dashboard interaction. Proper data modeling and optimized DAX calculations help improve report performance and user experience.

4.1 Reusability

The solution is designed to be reusable with updated sales datasets. The same data cleaning workflow, Power BI data model, and dashboard structure can be applied to future sales data without requiring major modifications. This reduces development time and improves scalability.

4.2 Application Compatibility

The project is compatible with Power BI Desktop for dashboard development and Power BI Service for online access and sharing. The dashboard can be viewed on desktops and laptops, allowing users to access business insights from different environments.

4.3 Resource Utilization

The project utilizes system resources efficiently through optimized data transformation, proper relationship management, and calculated measures. The use of Power Query and DAX helps minimize unnecessary calculations, improving report performance and reducing memory consumption. s

4.4 Deployment

The final dashboard can be deployed through Power BI Service, enabling online access and collaboration. Users can securely share reports with stakeholders, allowing management to monitor sales performance and business metrics in real time.

5. Dashboards

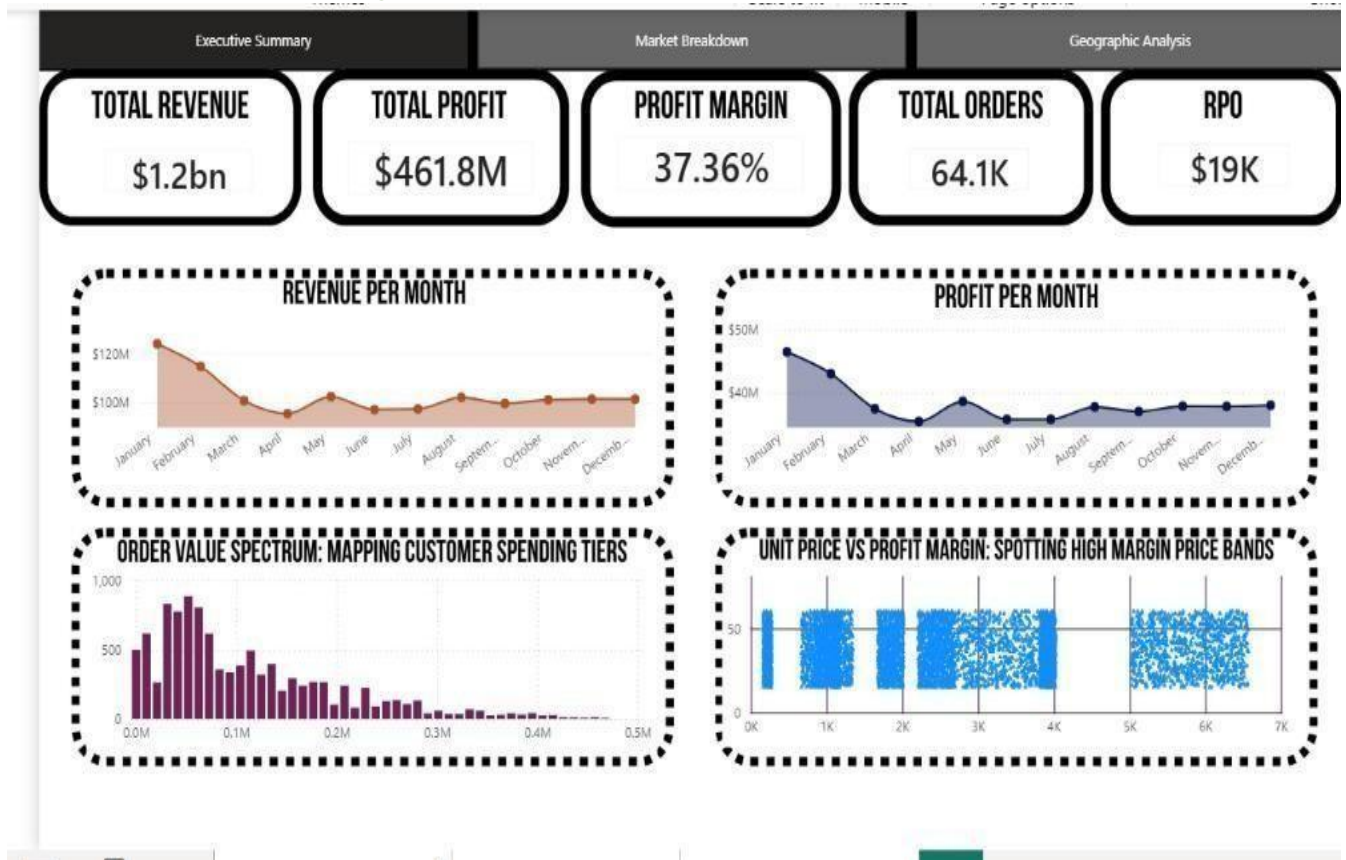
The dashboard presents sales data through interactive visualizations that help users understand revenue trends, profit performance, customer behavior, product performance, and regional sales distribution. Multiple dashboard pages were created to provide different analytical perspectives and support business decision-making.

5.1 KPIs (Key Performance Indicators)

The key performance indicators used in this project include:

- Total Revenue
- Total Profit
- Profit Margin
- Total Orders
- Revenue Per Order (RPO)
- Monthly Revenue Trend
- Monthly Profit Trend
- Regional Revenue Contribution
- Channel-wise Revenue Distribution
- Product Performance Analysis

5.2 Executive Summary Dashboard



Description:

This dashboard provides an overview of business performance through KPIs such as Revenue, Profit, Orders, and Profit Margin. It also includes monthly revenue and profit trends.

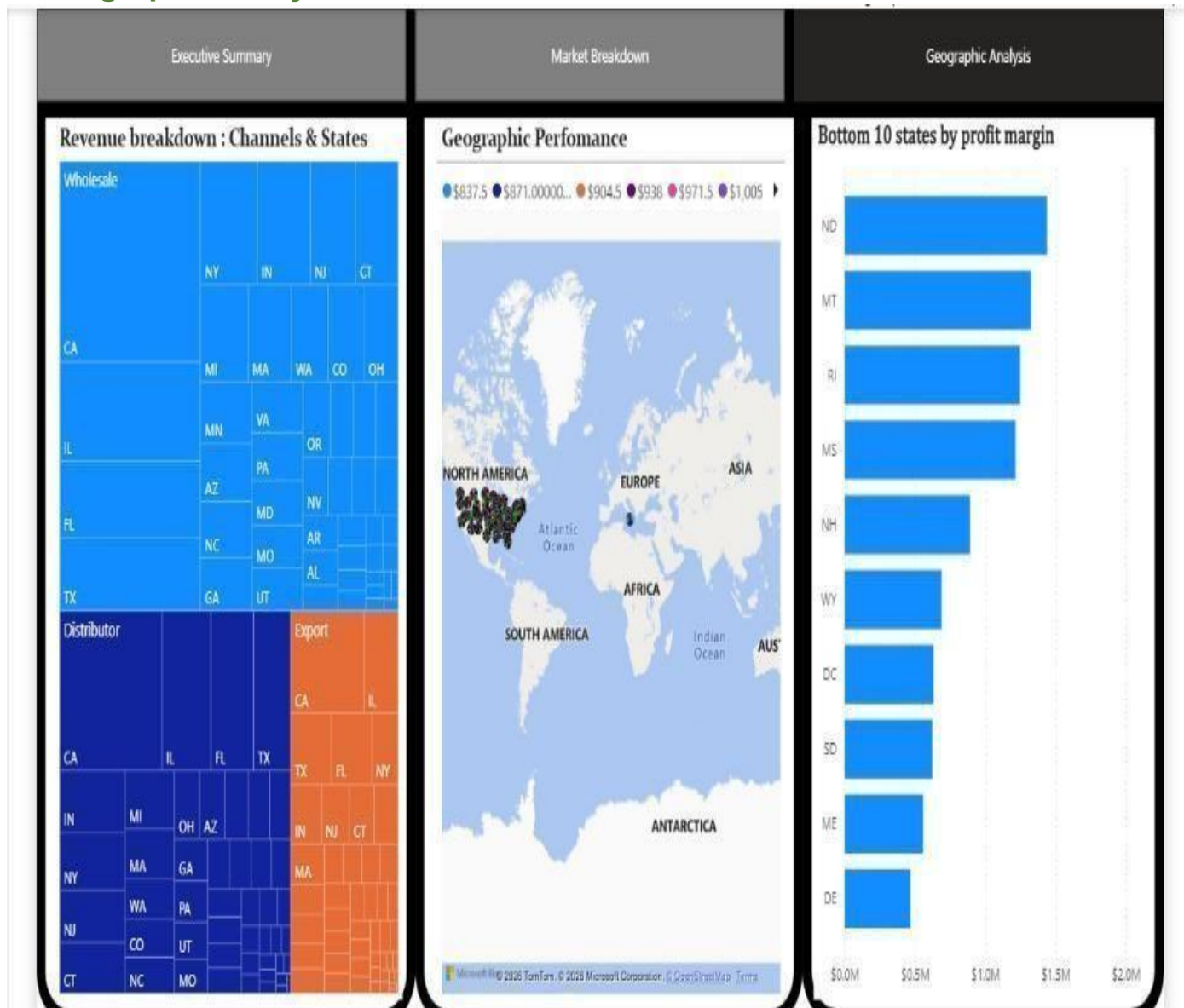
5.3 Market Breakdown Dashboard



Description:

This dashboard focuses on product performance, customer analysis, regional revenue contribution, and sales channel distribution.

5.4 Geographic Analysis Dashboard



Description:

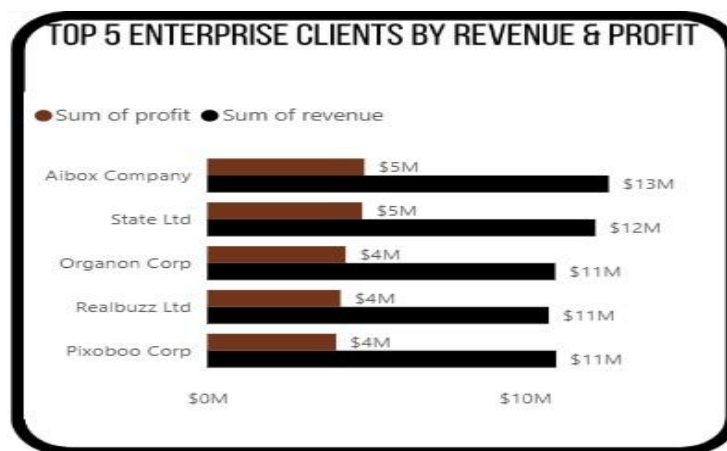
This dashboard presents state-wise sales performance using maps and regional visualizations to identify high-performing and low-performing areas.

5.5 Additional Findings

5.5.1 Customer Concentration (Top 5 Accounts)

The analysis revealed that a significant portion of the company's revenue and profit is generated by a small number of enterprise customers. These customers contribute a high volume of orders and play a critical role in the overall business performance. This indicates that maintaining strong relationships with these key accounts is essential for sustaining revenue growth and profitability.

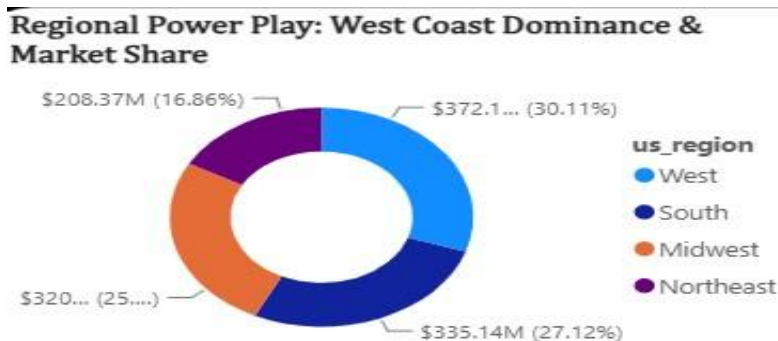
Customer Revenue Profit Orders



5.5.2 Macro Region Performance

To better understand geographic performance, sales data was aggregated into broader US regions. The results show that the West region dominates the company's revenue and profit generation, while the Northeast contributes the least among all regions.

Region Revenue Profit

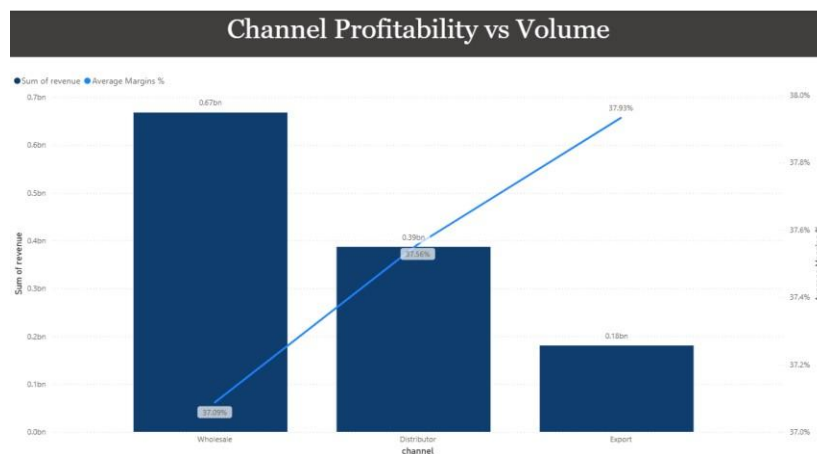


These findings indicate that the West region is the company's strongest market, while the Northeast region presents opportunities for future business expansion and improvement.

5.5.3 Top 5 States by Revenue

A state-level revenue analysis identified California as the highest-performing state by a significant margin. California alone generated more than double the revenue of the next highest-performing state, highlighting its importance to the company's overall sales performance.

State Revenue

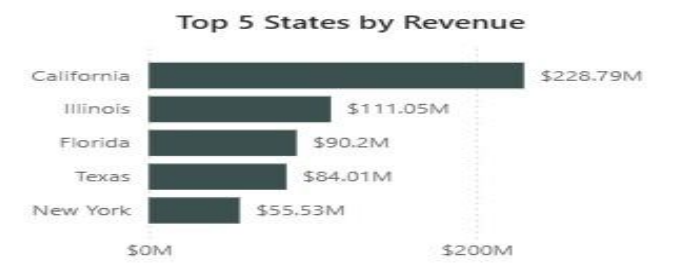


This analysis helps identify key geographic markets that contribute most significantly to business growth and revenue generation.

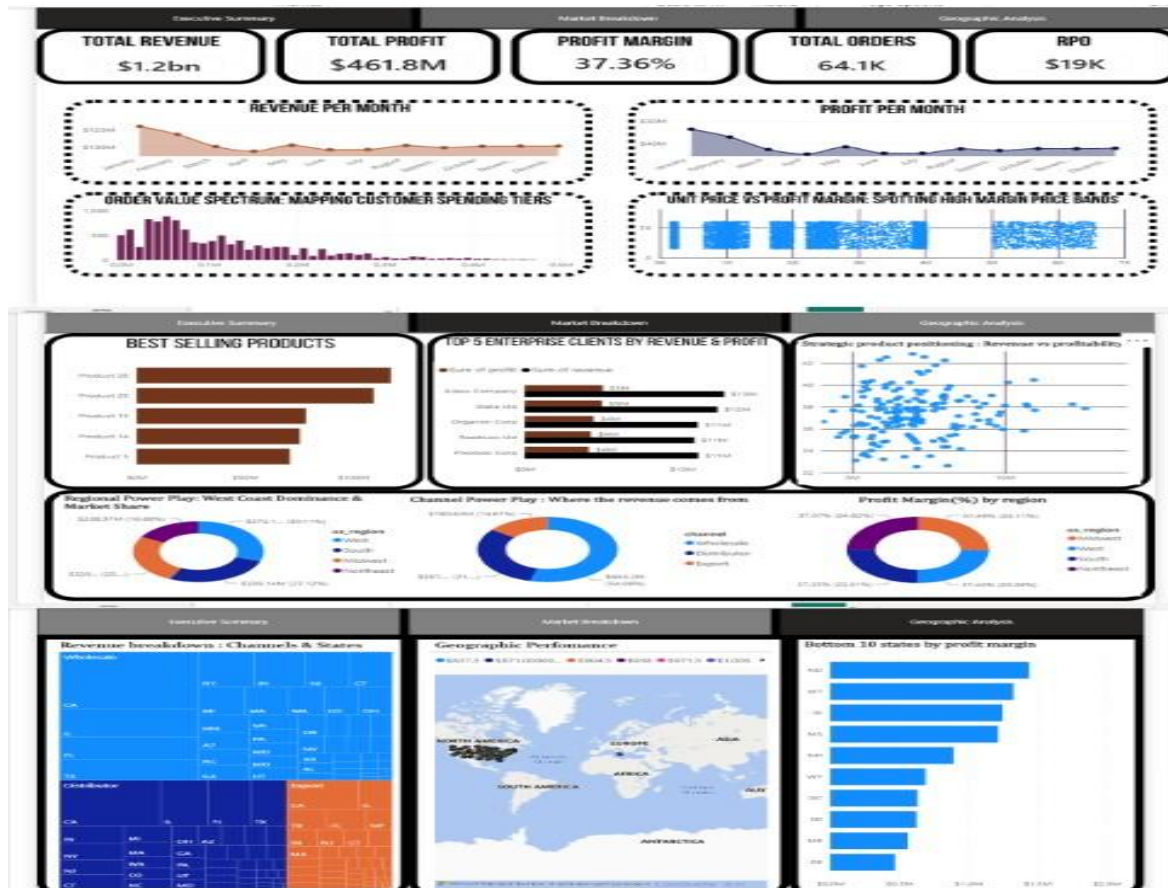
5.5.4 Channel Profitability vs Volume

While the Wholesale channel generated the highest overall sales volume, further analysis revealed differences in profitability across sales channels. The Export channel, despite generating the lowest total revenue, achieved the highest average profit margin, indicating better profitability per dollar sold.

Channel Revenue Profit Margin



5.6 Final Dashboard



Description:

The Final Dashboard combines the most important insights from all previous dashboard pages into a single consolidated view. It presents key performance indicators, revenue and profit trends, regional performance, customer analysis, product performance, and sales channel contributions. This dashboard serves as a summary view for management, enabling quick monitoring of overall business performance and supporting data-driven decision-making.

6. Conclusion

This project successfully analyzed regional sales data using Python, Google Colab, Microsoft Excel, and Power BI. The dataset, obtained from Kaggle's public platform, was cleaned, transformed, and analyzed to generate meaningful business insights. Through feature engineering and dashboard development, key metrics such as Revenue, Profit, Profit Margin, and Regional Performance were evaluated. The interactive dashboards helped visualize sales trends, product performance, customer behavior, and regional contributions. Overall, the project demonstrates how data analytics and business intelligence tools can transform raw data into valuable insights that support informed business decision-making and performance improvement.